Air Quality and Health Impacts Associated with Prescribed Fire and Agricultural Burning

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Background

- The recent rise in frequency and severity of wildfires increased the interest in the use of prescribed burn as a management tool nationwide.
- Prescribed burns decrease the buildup of flammable debris, vegetation and subsequent fuel for wildfires, mitigating the spread and intensity of wildfires.
- These burns are conducted under different environmental conditions than wildfire weather, leading to incomplete combustion.
- As a result, their smoke differs from wildfire smoke in terms of constituents, concentrations, and proportions.
- In addition, prescribed fire and agricultural burning can expose communities residing near burn areas to high levels of smoke.
- To mitigate population-level exposure, public health practitioners and decision-makers need a better understanding of the health impacts associated with smoke exposure from prescribed fire and agricultural burning.

Objectives

Aim 1: Estimate the air quality impacts associated with prescribed burns and exposure to both daily 24-hour average fine particulate matter ($PM_{2.5}$) and daily maximum 8-hour average ozone (O_3) levels in the Southeastern United States (US). *Hypothesis 1:* Prescribed and agricultural burning contribute to a large proportion of daily $PM_{2.5}$ and O_3 concentrations in the Southeastern US.

Aim 2: Quantify the effects of $PM_{2.5}$ and O_3 attributable to prescribed burns on rates of emergency room (ER) visits (all cause and cause-specific) and other healthcare utilizations in the Southeastern US. *Hypothesis 2*: Air pollution from prescribed burns is associated with a risk of adverse health outcomes.

Aim 3: Assess how the observed health impacts vary across strata defined by age, sex, and markers of socioeconomic and social vulnerability.

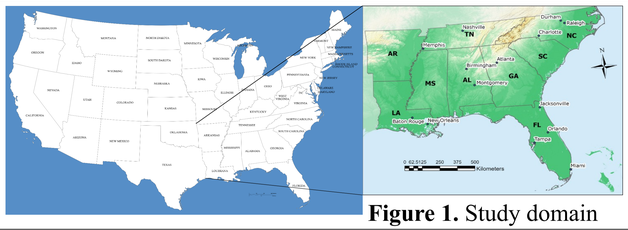
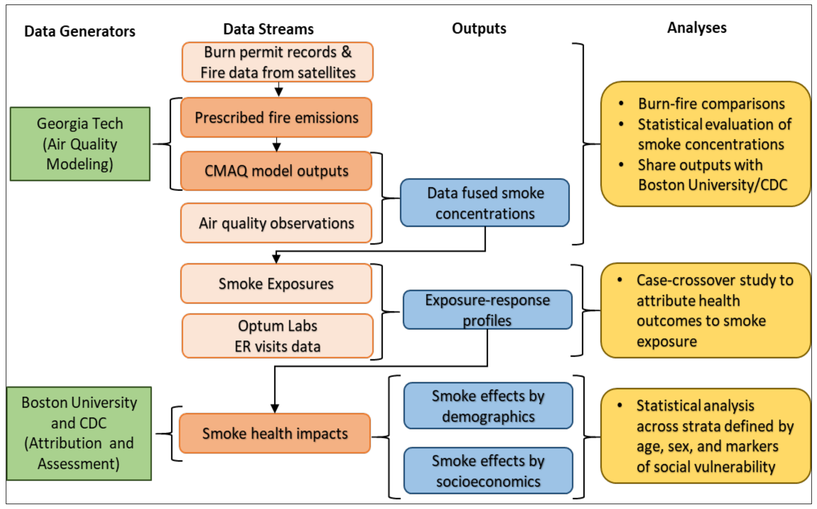


Figure 1. Study domain

Experimental Plan and Methods



- Emissions from permit or satellite-based data can be estimated using *Equation 1*. The emissions of a chemical species, *E*, is calculated by multiplying the fuel load consumed (*FC*, mass of burned fuel type per unit area) by the total area burned, *A*, and an emission factor (*EF*).

$$E = FC * A * EF \qquad (1)$$

The areas *A* can be derived from burn permits or satellite observations.

- BlueSky framework is being used to calculate fire emissions. BlueSky framework has *EFs* of 276 known air pollutants.
- Fire Emissions will be input to CMAQ to simulate prescribed burn impacts in the southeastern US during the burn season (November–April).
- The meteorological inputs will be taken from operational daily weather forecast simulations over the same domains with the WRF model.
- The effects of smoke attributable to prescribed burns will be estimated using health care administrative claims data.

Preliminary Results

1. Comparison of satellite-based fire data with burn permit records:

- The Fire INventory from NCAR (FINN) data includes all fires detected by MODIS and VIIRS satellites while permits are only for prescribed burning. So, for matching FINN data with permit data one needs to remove wildfires from the FINN dataset.
- For matching the burned areas, we conduct three approaches: (a) statewide matching (Figure 2), (b) fire to fire matching and (c) a grid-based matching.

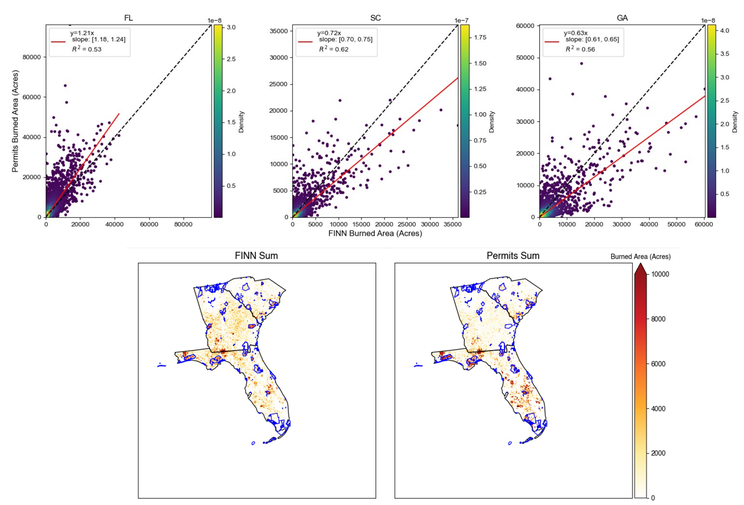


Figure 2. (top) Daily total burned area in (a) Florida (b) South Carolina and (c) Georgia, and (bottom) burned areas in those three states. Georgia burned area is the sum during 2015–2020 while Florida and South Carolina burned areas are sums of 2013–2020.

- ❑ A spatial and temporal clustering algorithm can detect the fires which have more than one day duration and identify them as wildfires.
- ❑ Prescribed burnings typically start and finish in the same day.
- ❑ Agricultural burnings can be detected using a Land Cover Classifications matching model.

Acknowledgement and Contact Information

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- For more information contact Talat Odman (*odman@gatech.edu*).

2. Quantifying the contributions of prescribed and agricultural burning:

- The daily $PM_{2.5}$ and O_3 concentrations as well as the contributions of the prescribed and agricultural burns to these metrics are estimated (Figure 3).
- Prescribed fire impacts on $PM_{2.5}$, O_3, NO_2, and CO concentrations over Georgia for the burning seasons (January–April) of 2015–2018 have been prepared.
- We mark the burn locations on these maps to show the associations between prescribed fires and smoke plumes.
- Daytime O_3 concentrations increase simultaneously with the increase in $PM_{2.5}$ due to the fire impact.

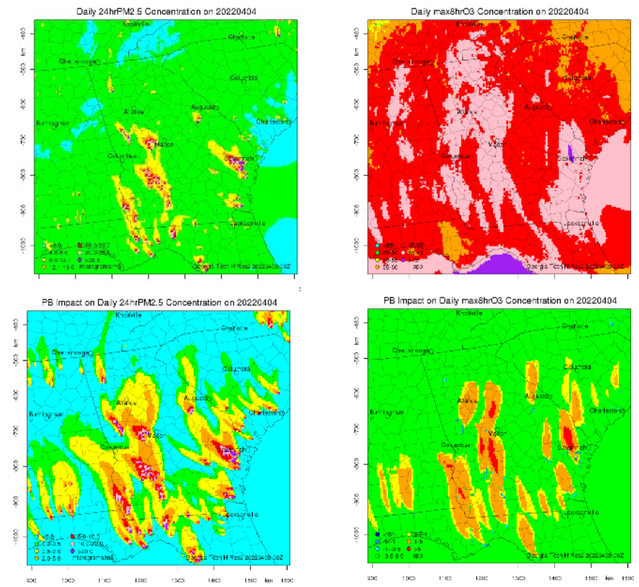
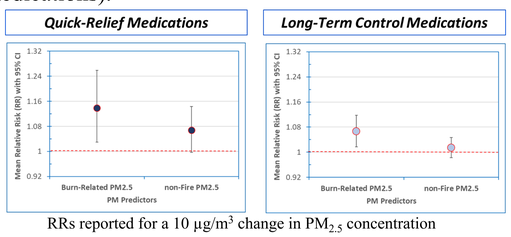


Figure 3. (top) Daily $PM_{2.5}$ and O_3 concentrations and (bottom) prescribed burn impacts on those concentrations during a burn day in the southeastern US.

3. The association between burn-related smoke-exposure and medication fills in Georgia during the active burning season:

- Primarily we evaluated associations between $PM_{2.5}$ concentrations from prescribed fires (burn $PM_{2.5}$) and various asthma medications, using state-of-the-art epidemiological regression techniques.
- Delayed effects of initial smoke exposure persisted for several days, and effect sizes varied depending on the type of medication (quick-relief vs. long-term controller medications).



RRs reported for a 10 μg/m³ change in $PM_{2.5}$ concentration

Figure 4. Associations between $PM_{2.5}$ predictors and asthma medication fills, over a cumulative 5-day lag period.

Conclusions

- A new spatial clustering algorithm is applied to FINN fire records to distinguish prescribed fires from wildfires.
- Mapping and temporal analysis of burn impacts and total concentrations show that the impacts are largest on burn days in proximity to the fires.
- Prescribed fire smoke exposure is associated with asthma medication fills in Georgia during the burn season.